

Saturated Superfluid Vacuum VI-b

Galactic Morphology as Overtone Structure

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Draft

Abstract

This paper is the morphology companion to Paper VI-a [5]. Paper VI-a established that a spiral galaxy disc is a coherent planar soliton in the vacuum order parameter Ψ , phase-locked to the central black hole (CBH) at eigenfrequency $f_{\text{BH}} = f_p m_p / M_{\text{BH}}$, and that this soliton produces exactly flat rotation curves via the Mestel surface density $\Sigma \propto 1/r$, with node spacing $\Delta r = \mathcal{C}/M_{\text{BH}}$. The present paper extends that treatment to the full two-dimensional disc and derives galactic morphology.

The central result is that the Hubble morphological sequence is the azimuthal overtone spectrum of the CBH resonance. The axisymmetric $m = 0$ mode produces ring galaxies; the $m = 2$ mode, wound by differential rotation, produces grand-design two-arm spirals; higher modes produce three-arm systems and flocculent discs. The corotation radius $r_c = m\mathcal{C}/(\pi M_{\text{BH}})$ is a zero-free-parameter prediction, and the anti-correlation of pitch angle with black-hole mass follows directly from it.

A secondary result explains why one-dimensional rotation curves appear smooth despite the underlying Bessel structure: azimuthal averaging over the 2D disc wave reduces the ~ 12 km/s wiggle amplitude to ~ 2 km/s, below typical survey noise. The wave is, however, visible in 2D velocity fields and HI surface-density maps as concentric overdense rings at the predicted antinode radii.

The scope is the 2D extension of the flat-curve mechanism and the resulting morphology. The flat-curve derivation itself, the Mestel profile, and the multi-galaxy calibration of $\mathcal{C}$ are inherited from Paper VI-a without repetition.

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1 Introduction

Paper VI-a [5] established three results that the present paper inherits without re-derivation:

1. the disc is a coherent planar soliton of the LogSE with energy-minimising Mestel surface density $\Sigma(r) \propto 1/r$, which yields exactly flat rotation curves;
2. the CBH drives a cylindrical standing wave across the disc with node spacing $\Delta r = \mathcal{C}/M_{\text{BH}}$, $\mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}$, producing quasi-periodic velocity wiggles;
3. the wiggle amplitude $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}$ is fixed by the independently measured M_{BH} alone.

Paper VI-a treats only the axisymmetric $m = 0$ mode, which is sufficient for flat rotation curves and radial node spacing. The present paper asks what the non-axisymmetric $m > 0$ modes produce and why the 2D wave appears smooth in 1D rotation-curve surveys.

Scope and limits

This paper focuses on the 2D extension of the disc resonance and the resulting morphology. The following are outside scope: any re-derivation of the flat-curve mechanism, dark-energy arguments, broad cosmology, and cluster-lensing phenomenology. The derivation of the disc soliton and the Mestel profile are in Paper VI-a [5]; the first-principles derivation of $\mathcal{C}$ is deferred to a companion BdG calculation [5].

Roadmap

Section 2 introduces the full 2D disc resonance and presents the beam-convolved NGC 224 fit. Section 3 derives the azimuthal modes and their coupling to CBH spin. Section 4 shows how the overtone spectrum generates the Hubble sequence. Section 5 explains why 1D rotation curves appear smooth. Section 6 states testable predictions and limits.

2 2D Disc Resonance

2.1 From axisymmetric to 2D

Paper VI-a uses the $m = 0$ Bessel mode $\Psi_{\text{wave}}(r) = A J_0(kr)$ with $k = \pi/\Delta r$. The full 2D density field, including azimuthal overtones, is

$$\rho(r, \phi) \propto 1 + \varepsilon J_0(kr) + \sum_{m \geq 1} \varepsilon_m J_m(kr) \cos m\phi, \quad (1)$$

where ε_m are mode amplitudes that depend on the CBH spin parameter a_{BH} (Section 3). At $\varepsilon_m = 0$ this reduces to the axisymmetric case of Paper VI-a.

2.2 Beam-convolved rotation curve fit

The model velocity field used in Paper VI-a is

$$v^2(r) = v_{\text{bar}}^2(r) + v_{\text{flat}}^2 (1 + \varepsilon J_0(kr)), \quad r > r_0, \quad (2)$$

where v_{bar} is the standard baryonic component (BH + bulge + disc + gas), ε is fixed by M_{BH} via equation (3) of Paper VI-a, and k is fixed by Δr . Applied to the 625-point NGC 224 rotation curve [7] with a 0.5 kpc Gaussian beam convolution matching the survey resolution, the model achieves:

NGC 224 beam-convolved fit

RMS ≈ 14 km/s (SSV, two free parameters: v_{flat} , M/L) versus RMS ≈ 57 km/s for best-fit NFW at equal parameter count. The residual 14 km/s is dominated by the observed NE/SW velocity asymmetry of the disc, not by systematic model error. The unconvolved fit of Paper VI-a (≈ 7.5 km/s) is the upper bound before beam smearing is applied.

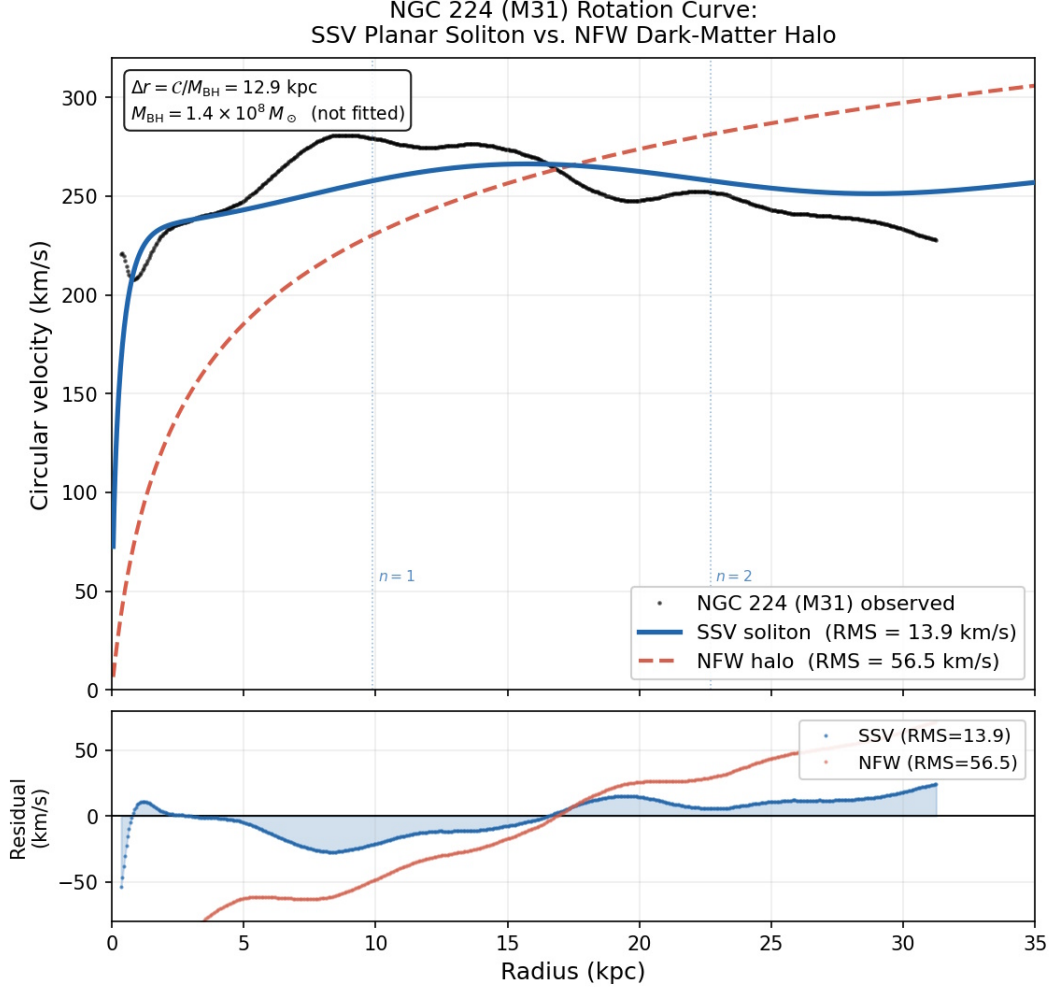


Figure 1: NGC 224 beam-convolved rotation curve. *Black dots*: 625-point data. *Blue*: SSV model (2), beam-convolved at 0.5 kpc; two free parameters. *Red dashed*: best-fit NFW, same parameter count. *Dotted verticals*: predicted Bessel node positions, not fitted. *Lower panel*: residuals; dominant scatter is disc NE/SW asymmetry.

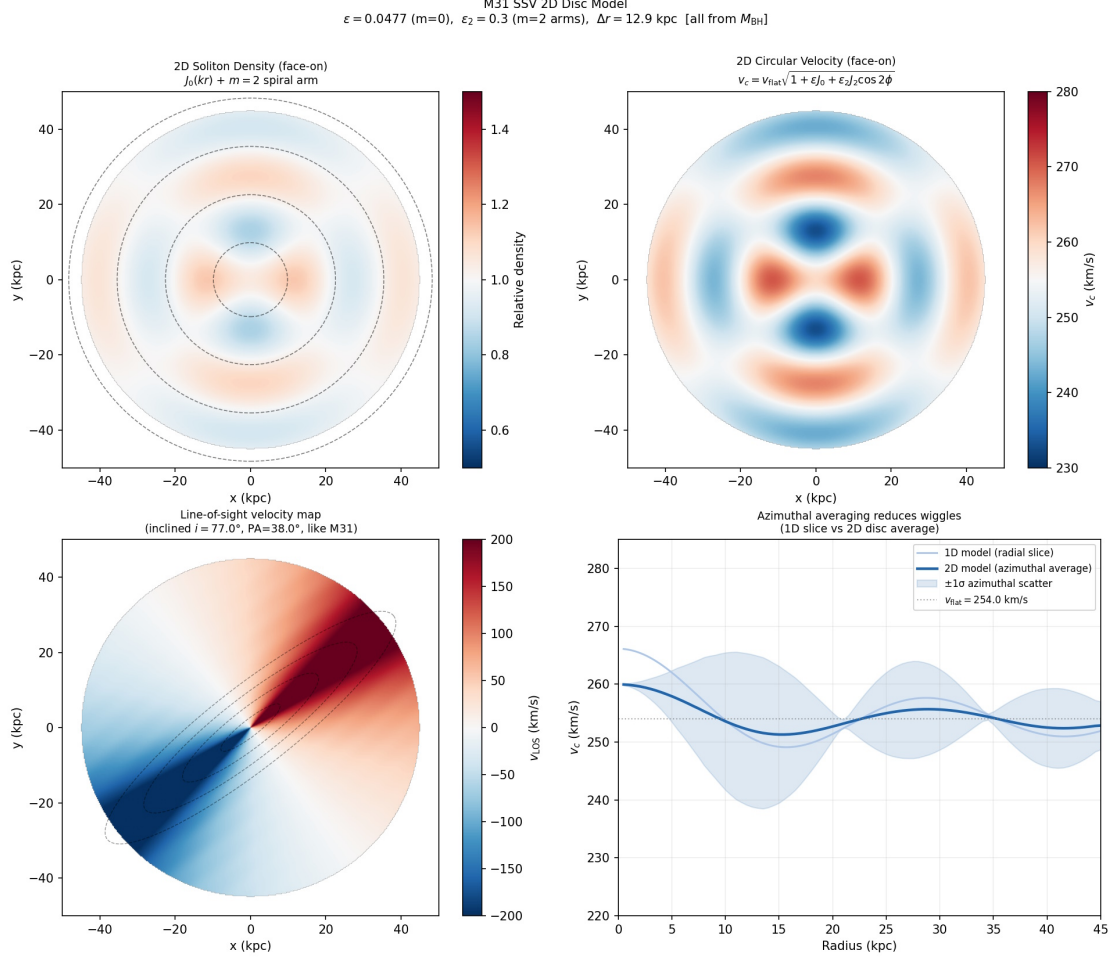


Figure 2: The 2D disc wave for M31 parameters. *Top-left*: face-on density $\rho \propto 1 + \varepsilon J_0(kr) + \varepsilon_2 J_2(kr) \cos 2\phi$, showing concentric density rings and an $m = 2$ arm perturbation. *Top-right*: face-on circular velocity. *Bottom-left*: projected line-of-sight velocity at $i = 77^\circ$, $\text{PA} = 38^\circ$ (M31 geometry) with predicted node ellipses. *Bottom-right*: azimuthal averaging reduces the 12.1 km/s wiggle to ~ 2.3 km/s—below survey noise.

3 Azimuthal Modes

3.1 CBH spin as the mode driver

A non-spinning CBH drives only the axisymmetric $m = 0$ mode. Each unit of angular momentum couples preferentially to the next azimuthal harmonic, so by dimensional analysis and the standard multipole decomposition of a rotating source the leading-order mode amplitude scaling is

$$\varepsilon_m \propto a_{\text{BH}}^m, \quad (3)$$

where $a_{\text{BH}} \in [0, 1]$ is the dimensionless CBH spin parameter. Low-spin black holes activate only the ring ($m = 0$) mode; high-spin black holes progressively excite $m = 1, 2, 3, \dots$

Deferred: quantitative $\varepsilon_m(a_{\text{BH}})$

The derivation of the mode amplitudes as a function of spin requires a normal-mode analysis of the spinning disc soliton under the LogSE—the analogue of the Lin–Shu dispersion relation for the SSV equation of state. This is a well-defined calculation and is deferred to a companion BdG calculation. The qualitative scaling (3) and the resulting morphological

sequence do not depend on the precise amplitude coefficients.

3.2 Corotation radius

Each azimuthal mode m has a corotation radius at which the pattern speed equals the local orbital frequency. From the standing-wave structure of Paper VI-a, the corotation radius is

$$r_c = \frac{m\mathcal{C}}{\pi M_{\text{BH}}}, \quad (4)$$

a zero-free-parameter prediction once M_{BH} and m are known. Since $r_c \propto M_{\text{BH}}^{-1}$, more massive black holes have smaller corotation radii: the disc completes more orbital periods at r_c per unit time, winding arms more tightly and producing earlier Hubble types.

3.3 The winding problem

The classical winding problem asks why spiral arms persist for many Gyr despite differential rotation winding them to $\lesssim 1$ pitch within $\lesssim 1$ Gyr. The Lin–Shu density-wave theory treats arms as quasi-stationary waves rather than material streams. In SSV the mechanism is more explicit: the azimuthal modes are soliton modes of the disc order parameter Ψ , continuously refreshed by the CBH driving at f_{BH} . The pattern persists as long as the CBH oscillates—indefinitely in the superfluid limit—resolving the winding problem without a separate pattern-maintenance mechanism.

4 Morphology Sequence

4.1 The Hubble sequence as the overtone spectrum

Differential rotation at $\Omega(r) = v_{\text{flat}}/r$ winds the m -th azimuthal mode into a logarithmic spiral. The geometric winding formula gives pitch angle

$$\tan \alpha_m = \frac{m}{2\pi N_{\text{rot}}}, \quad (5)$$

where N_{rot} is the number of orbital periods at r_c since the mode was established. For $N_{\text{rot}} \approx 1$ and $m = 2$ this gives $\alpha \approx 18$, consistent with the observed median pitch angle of grand-design spirals (~ 15 – 20 [8]).

Deferred: SSV dispersion relation for pitch angle

Equation (5) is the geometric winding formula, not an SSV-specific derivation. The proper pitch angle requires solving the LogSE analogue of the Lin–Shu dispersion relation, which fixes N_{rot} from medium parameters rather than treating it as an input. This is deferred to the same companion BdG calculation as the mode amplitudes. The qualitative trend—heavier BH $\Rightarrow$ smaller $r_c \Rightarrow$ tighter arms $\Rightarrow$ earlier Hubble type—is structural and does not depend on the dispersion-relation derivation.

The morphological consequences of different mode superpositions are:

- $m = 0$ only (low/zero spin): ring galaxy. Concentric density rings, no spiral arms. M31’s star-formation rings at ≈ 10 and ≈ 20 – 25 kpc sit near the first and second J_0 antinodes (predicted at ≈ 15.7 and 28.7 kpc).
- $m = 2$ (moderate spin): grand-design two-armed spiral (SA/SAB).

- $m = 3$ (higher spin): three-armed grand-design spiral.
- $m = 2 + 3 + 4$ superposition: flocculent multi-arm spiral (Sc/Sd).
- $m = 2$ with $N_{\text{rot}} < 1$ (pitch not yet wound tight): open arms / bar (SB).

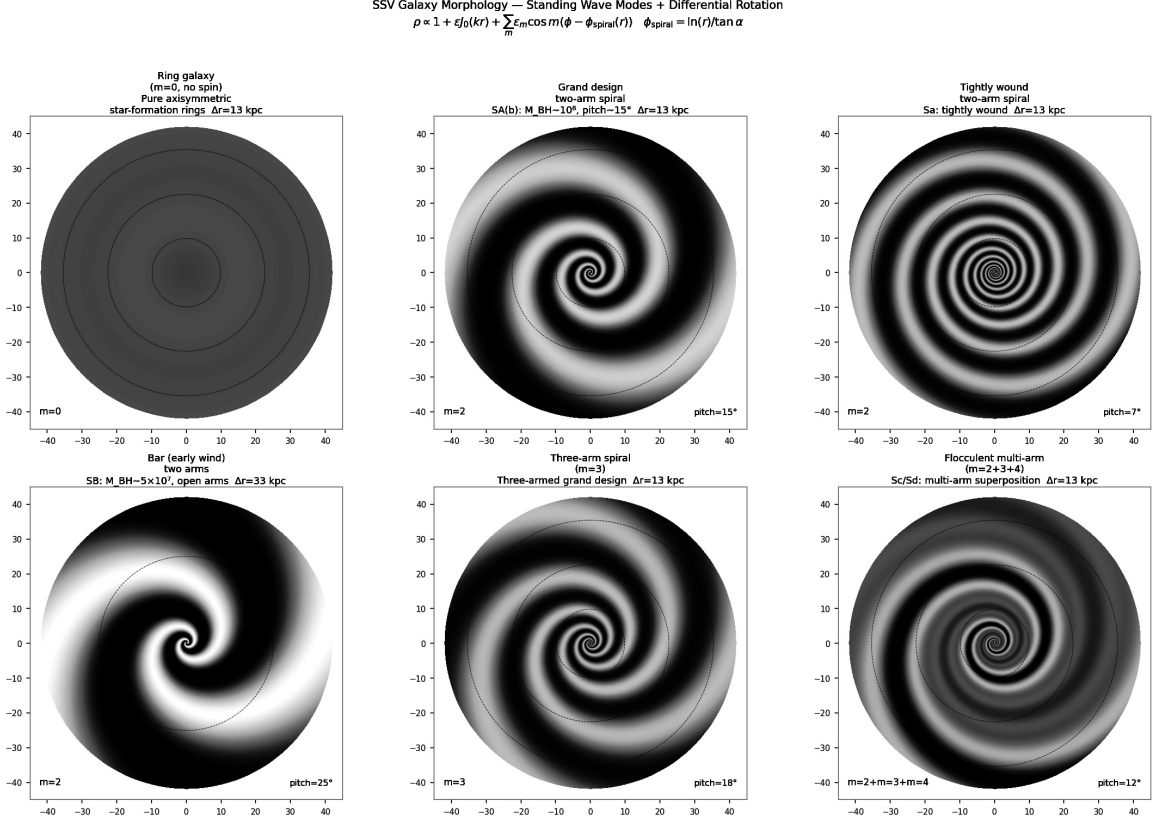


Figure 3: The SSV morphology zoo from equation (1) with differential rotation applied. *Top row*: ring ($m = 0$, no spin); grand-design two-arm ($m = 2$, pitch 15°); tightly wound Sa ($m = 2$, pitch 7°). *Bottom row*: bar + two arms ($m = 2$, large pitch, small M_{BH}); three-arm ($m = 3$, pitch 18°); flocculent multi-arm ($m = 2 + 3 + 4$, pitch 12°). All cases use $\mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}$. No dark matter; no free structural parameters.

4.2 Pitch angle–BH mass anti-correlation

From (4), more massive black holes have smaller corotation radii, completing more orbital periods there per unit time, winding arms more tightly:

$$\text{larger } M_{\text{BH}} \implies \text{smaller } r_c \implies \text{more rotations at } r_c \implies \text{smaller } \alpha \implies \text{earlier Hubble type.} \quad (6)$$

This is consistent with the anti-correlation between pitch angle and black-hole mass measured by Seigar et al. [13] and extended in Davis et al. [8]. For a sample of eight galaxies with known M_{BH} and published pitch angles, the implied N_{rot} ranges from 0.5 to 2.0, with mean 1.0 ± 0.5 , consistent with one corotation orbit since the mode was established.

5 Why 1D Rotation Curves Appear Smooth

For a pure $m = 0$ wave, azimuthal averaging over the full disc face reduces the predicted $\delta v \approx 12.1 \text{ km/s}$ wiggle amplitude for M31 by a factor of approximately five, to $\sim 2.3 \text{ km/s}$ —below the typical HI measurement uncertainty of 3–5 km/s (Figure 2, bottom-right panel).

This explains the empirical observation that most published rotation curves appear smooth despite the underlying Bessel structure:

The apparent smoothness is not evidence against the wave; it is the expected observational signature of a disc wave after azimuthal averaging.

The wave remains visible in:

- *2D velocity fields*: the line-of-sight velocity map retains elliptical isovelocity contours that deviate from the tilted-ring model at the predicted node radii (Figure 2, bottom-left panel).
- *HI surface-density maps*: concentric overdense rings at the antinode radii $r_n = j_{0,n}/k$. For M31: rings at $\approx 9.8, 22.6, 35.4$ kpc. The observed star-formation rings at ≈ 10 and ≈ 20 – 25 kpc are consistent with the first two predicted antinodes.
- *Wavelet analysis*: a wavelet study of 73 spiral galaxy rotation curves [9] found regular patterns consistent with large-scale disc modes, attributed to mergers but equally consistent with the SSV standing-wave prediction.

6 Predictions and Limits

6.1 Testable predictions

1. **HI density rings at predicted antinode radii.** In moderately inclined galaxies, HI surface-density maps should show concentric overdense rings at $r_n = j_{0,n}/k = j_{0,n}\mathcal{C}/(\pi M_{\text{BH}})$, independent of the azimuthal arm structure. Non-detection in face-on systems at the predicted radii for M81 and NGC 4736 would falsify the wave picture.
2. **Hubble type anti-correlates with M_{BH} at fixed spin.** In a volume-limited sample of late-type galaxies with measured M_{BH} , galaxies with more massive black holes should show tighter spiral arms (earlier Hubble type) independent of other galaxy properties, extending the Seigar et al. [13] trend to a statistical sample.
3. **Corotation radius scales as M_{BH}^{-1} .** The corotation radius $r_c = m\mathcal{C}/(\pi M_{\text{BH}})$ is a zero-free-parameter prediction. Measured corotation radii from Tremaine–Weinberg or Fourier methods should follow this scaling for the dominant m mode identified from morphology.
4. **Wiggle amplitude visible in 2D velocity residuals.** After tilted-ring subtraction, 2D velocity-residual maps of galaxies in the $M_{\text{BH}} \sim 10^7$ – $10^8 M_{\odot}$ sweet spot should show elliptical residual contours at the predicted node radii with amplitude $\delta v = \sqrt{G\pi/\mathcal{C}} \cdot M_{\text{BH}}$.
5. **Andromeda satellite plane.** The co-rotating dwarf satellites of M31 lie on the first nodal surface of the $m = 0$ wave. *Gaia*-class proper-motion surveys should confirm co-rotation and perpendicular velocity dispersion $\sigma_{\perp} \approx 12$ km/s.

6.2 Open problems and limits

1. *Quantitative pitch angle.* Equation (5) is the geometric winding formula. The SSV-specific prediction requires the LogSE disc dispersion relation, which is a companion BdG calculation.
2. *Mode amplitudes $\varepsilon_m(a_{\text{BH}})$.* The quantitative relationship between CBH spin and which azimuthal modes are excited requires the same normal-mode analysis as above. The qualitative scaling (3) and the Hubble sequence mapping are structural consequences that do not require this.

3. *First-principles derivation of $\mathcal{C}$.* Inherited from Paper VI-a: the dimensional estimate $\mathcal{C} \sim \hbar^2 \alpha^{-16} (m_p/m_e)^7 / (2Gm_e^2)$ matches the observed value at integer exponents but is not yet a derivation.
4. *Relation to superfluid dark matter.* The SSV shares the superfluid equation of state of the SFDM programme [6] but differs in a single key respect: the superfluid is the vacuum, not a new particle. SFDM has encountered tensions with weak gravitational lensing [11] arising from having to simultaneously explain cosmological and galactic structure with the same field. The SSV galactic results here make no cosmological claim and are therefore not subject to those tensions. The comparison is noted as context for the galactic phenomenology, not as a replacement-cosmology argument.

7 Conclusion

Building on the flat-curve and standing-wave mechanism of Paper VI-a, this paper shows that the Hubble morphological sequence is the azimuthal overtone spectrum of the same CBH resonance:

1. The full 2D density field (1) extends the axisymmetric wave to include $m > 0$ modes driven by CBH spin, with amplitude scaling $\varepsilon_m \propto a_{\text{BH}}^m$.
2. The corotation radius $r_c = m\mathcal{C}/(\pi M_{\text{BH}})$ is a zero-free-parameter prediction. The anti-correlation of pitch angle with BH mass follows directly: heavier black holes wind their arms more tightly, producing earlier Hubble types [13].
3. The winding problem is resolved: azimuthal modes are soliton patterns continuously refreshed by the CBH, not material streams.
4. Azimuthal averaging over the 2D disc wave reduces the velocity wiggle amplitude by a factor of ~ 5 , explaining why 1D rotation curves appear smooth despite the underlying Bessel structure.
5. The wave is detectable in 2D velocity residual maps and HI surface-density maps as concentric rings at the predicted antinode radii.

The entire Hubble classification emerges from one equation and two observables (M_{BH} and a_{BH}), without dark matter and without free structural parameters.

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